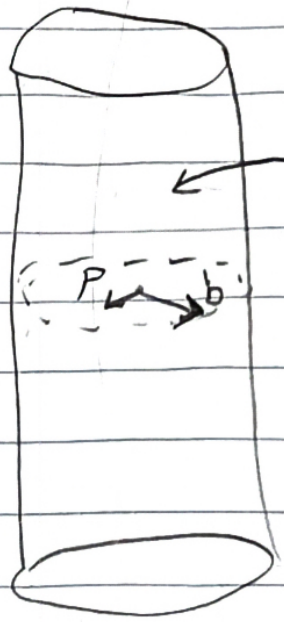
1.)

objective:

Prove:

- $\Phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} \phi(b,\phi') \frac{b^2-\rho^2}{b^2+\rho^2-2b\rho\cos(\phi-\phi')} d\phi'$
- $\nabla^2\Phi = -\frac{\rho}{\varepsilon_0}$, $\nabla^2\Phi = 0$



Boundary Conditions:

- $\Phi(\rho,\phi=0) = \Phi(\rho,\phi=2\pi)$
- $\Phi(\rho=0,\phi) =$ finite value
- $\Phi(\rho=b,\phi) = V(\phi)$

(oordinates (ρ, φ)

- $\nabla^2 = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2}{\partial \phi^2}$
- $\nabla^2 \Phi = \frac{1}{\rho}\frac{\partial}{\partial \rho}\left(\rho \frac{\partial \Phi}{\partial \rho}\right) + \frac{1}{\rho^2}\frac{\partial^2 \Phi}{\partial \phi^2}$
  - $\Phi(\rho, \phi) = R(\rho)\,\Psi(\phi)$
  - multiply $\frac{\rho^2}{\Phi}$
- $\frac{\rho}{R}\frac{d}{d\rho}\left(\rho \frac{dR}{d\rho}\right) + \frac{1}{\Psi}\frac{\partial^2 \Psi}{\partial \phi^2} = 0$
  - $\frac{\rho}{R}\frac{d}{d\rho}\left(\rho \frac{dR}{d\rho}\right) = k^2$
  - $\frac{1}{\Psi}\frac{d^2\Psi}{d\phi^2} = -k^2$

$\Psi(\phi) = A\sin(k\phi + \alpha)$

- $\Psi(\phi) = A\cos(k\phi) + B\sin(k\phi)$

- $\frac{\rho}{R}\left(\rho \frac{d^2R}{d\rho^2} + \frac{dR}{d\rho}\right) = k^2$

- $\rho^2 \frac{d^2R}{d\rho^2} + \rho \frac{dR}{d\rho} = RK^2$

- $\rho^2 \frac{d^2R}{d\rho^2} + \rho \frac{dR}{d\rho} - RK^2 = 0$

$R = \rho^k$

$R' = k\rho^{k-1}$

$R'' = k(k-1)\rho^{k-2}$

- $k(k-1)\rho^2\rho^{k-2} + \rho k\rho^{k-1} - \rho^k k^2 = 0$
- $\rho^k[k(k-1) + k - k^2] = 0$

- $k^2 - k + k - k^2 = 0$

$0 = 0$

$R = \rho^{-k}$

$R' = -k\rho^{-k-1}$

$R'' = k(k+1)\rho^{-k-2}$

- $k(k+1)\rho^{-2}\rho^{-k} - k\rho\rho^{-1}\rho^{-k} - \rho k^2 = 0$

- $\rho^{-k}[k^2 + k - k - k^2] = 0$

$0 = 0$

- $R(\rho) = a\rho^{k} + b\rho^{-k}$

For small approximation $k=0$:

- $R(\rho) = a_0 + b_0 \ln(\rho)$
- $\Phi(\phi) = A_0 + B_0\phi$
- General solution = Small terms ($k \cong 0$) + Large terms ($k \geq 0$)
- Say ($k=n$)

$$\Phi(\rho,\phi) \; a_0 + b_0 \ln(\rho) + A_0 + B_0\phi + \sum_{n=1}^{\infty} A_n \sin(n\phi + \alpha_n)\rho^{n} + \sum_{n=1}^{\infty} B_n \rho^{-n} \sin(n\phi + \alpha_n)$$

- $(a_0 + A_0 = C_0)$

- I apply $\Phi(\rho=0,\phi)$ = Finite then all B-coefficients are invalid to B'CS.

- $\Phi(\rho,\phi) = C_0 + \sum_{n=1}^{\infty} A_n \rho^n \sin(n\phi + \alpha_n)$
- (Lets say that $\frac{C_n}{b} = A_n$)
- $\Phi(\rho,\phi) = C_0 + \sum_{n=1}^{\infty} C_n \left(\frac{\rho}{b}\right)^n \sin(n\phi + \alpha_n)$
  - $\sin\theta = \frac{e^{i\theta} - e^{-i\theta}}{2i}$
- $\Phi(\rho,\phi) = C_0 + \sum_{n=1}^{\infty} \frac{C_n}{2i} \left(\frac{\rho}{b}\right)^n \left[e^{i\alpha_n} e^{in\phi} - e^{-in\phi} e^{-i\alpha_n} \right]$

  ($e^{i\alpha_n} e^{in\phi}$: P_n; $e^{-in\phi} e^{-i\alpha_n}$: l_n)
- $P_n = \frac{e^{i\alpha_n}}{2i}$; $l_n = \frac{-e^{-i\alpha_n}}{2i}$
- $\Phi(\rho,\phi) = C_0 + \sum_{n=1}^{\infty} C_n \left(\frac{\rho}{b}\right)^n \left[P_n e^{in\phi} + l_n e^{-in\phi} \right]$

Including negative numbers.

- extend summation from $-\infty$ to ∞.

- $\Phi(\rho,\phi) = \sum_{n=-\infty}^{\infty} C_n \left(\frac{\rho}{b}\right)^{|n|} e^{in\phi}$ ← exponential fourier transform

apply $\Phi(\rho=b,\phi) = V(\phi)$

- $V(\phi) = \sum_{n=-\infty}^{\infty} C_n e^{in\phi}$

multiply both sides by $\int_0^{2\pi} e^{-in\phi}\, d\phi$

- $\int_0^{2\pi} V(\phi) e^{-in'\phi}\, d\phi = \sum_{n=-\infty}^{\infty} C_n \int_0^{2\pi} e^{i(n-n')\phi}\, d\phi$

- $\int_0^{2\pi} V(\phi) e^{-in\phi}\, d\phi = C_n 2\pi \delta_{nn'}$ $\quad n=n'$

$$C_n = \frac{1}{2\pi} \int_0^{2\pi} V(\phi') e^{-in\phi'}\, d\phi'$$

- $\phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} d\phi' V(\phi') \sum_{n=-\infty}^{\infty} \left(\frac{\rho}{b}\right)^{|n|} e^{in(\phi-\phi')}$

- $\Phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} d\phi' V(\phi') \left[-1 + \sum_{n=0}^{\infty}\left(\frac{\rho}{n}\right) e^{n\,in(\phi-\phi')}\right.$

$\left. + \sum_{n=0}^{\infty}\left(\frac{\rho}{b}\right) e^{n\,-in(\phi-\phi')}\right]$

* $\sum_0^{\infty} r^n = \frac{1}{1-r}$

$$\Phi(\rho,\phi) = \frac{1}{2\pi}\int_0^{2\pi} d\phi' V(\phi') \left[-1 + \frac{1}{1-\left(\frac{\rho}{b}\right)e^{i(\phi-\phi')}}\right.$$

$$\left. + \frac{1}{1-\left(\frac{\rho}{b}\right)e^{-i(\phi-\phi')}}\right] =$$

$$\left[\frac{-1+(\rho/b)e^{i(\phi-\phi')}}{1-\left(\frac{\rho}{b}\right)e^{i(\phi-\phi')}} + \frac{1}{1-\left(\frac{\rho}{b}\right)e^{i(\phi-\phi')}} + \frac{1}{1-\left(\frac{\rho}{b}\right)e^{-i(\phi-\phi')}}\right]$$

$$= \left[\frac{(\rho/b)e^{i(\phi-\phi')}}{1-(\rho/b)e^{i(\phi-\phi')}} + \frac{1}{\left(1-(\rho/b)e^{-i(\phi-\phi')}\right)}\right]$$

$$\cos\theta = \frac{e^{i\theta}+e^{-i\theta}}{2}$$

$$= \left[\frac{\left(\frac{\rho}{b}\right) e^{i(\phi-\phi')} \left(1-\left(\frac{\rho}{b}\right) e^{-i(\phi-\phi')}\right)}{\left(1-\left(\frac{\rho}{b}\right) e^{-i(\phi-\phi')}\right)\left(1-\left(\frac{\rho}{b}\right) e^{i(\phi-\phi')}\right)} \right.$$

$$\left. + \frac{\left(1-\left(\rho/b\right) e^{i(\phi-\phi')}\right)}{\left(1-\left(\frac{\rho}{b}\right) e^{i(\phi-\phi')}\right)\left(1-\left(\frac{\rho}{b}\right) e^{-i(\phi-\phi')}\right)} \right]$$

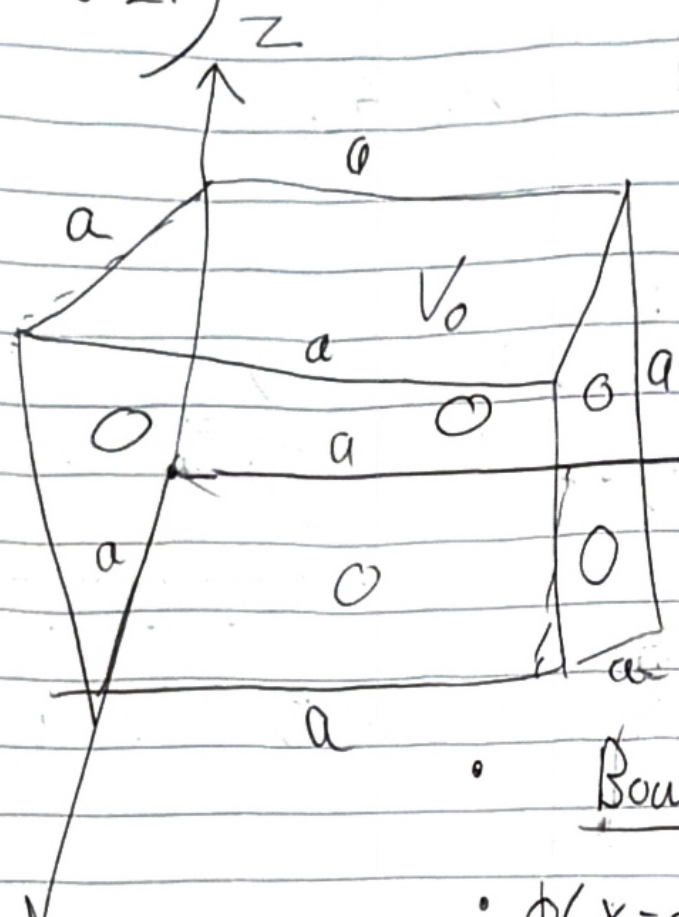
$$= \left[\frac{\frac{\rho}{b} e^{i(\phi-\phi')} - \frac{\rho^2}{b^2} + 1 - \frac{\rho}{b} e^{i(\phi-\phi')}}{1 - \frac{\rho}{b} e^{-i(\phi-\phi')} - \frac{\rho}{b} e^{i(\phi-\phi')} + \frac{\rho^2}{b^2}} \right]$$

$$= \left[\frac{-\rho^2 + b^2}{b^2 - 2b\rho\cos(\phi-\phi') + \rho^2} \right]$$

$$\phi(p=b, \phi') = V(\phi)$$

$$\phi(p, \phi) = \frac{1}{2\pi} \int_0^{2\pi} \phi(b, \phi') \left(\frac{b^2 - p^2}{b^2 + p^2 - 2bp\cos(\phi - \phi')} \right) d\phi'$$

Prob 2.)



(x, y, z) → coordinates

trial functions:

1.) sine & cosine
2.) exponential
3.) hyperbolic trig

Boundary Conditions:

- $\phi(x=0) = 0$
- $\phi(x=a) = 0$

use (1)

- $\phi(y=0) = 0$
- $\phi(y=a) = 0$

use (1)

- $\phi(z=0) = 0$
- $\phi(z=a) = V_0$

use (2)

- $\nabla^2\phi = 0$
- $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$
- $\vec{E} = -\nabla\phi$, $\int_S \vec{E}\cdot d\vec{A} = \frac{Q_V}{\varepsilon_0}$

$$\frac{\partial^2\phi}{\partial x^2} + \frac{\partial^2\phi}{\partial y^2} + \frac{\partial^2\phi}{\partial z^2} = 0$$

- $\phi(x,y,z) = A(x)B(y)C(z)$

- $\frac{1}{A}\frac{\partial^2 A}{\partial x^2} + \frac{1}{B}\frac{\partial^2 B}{\partial y^2} + \frac{1}{C}\frac{\partial^2 C}{\partial z^2} = 0$

  $\frac{1}{A}\frac{\partial^2 A}{\partial x^2} = -k^2$ $\quad \frac{1}{B}\frac{\partial^2 B}{\partial y^2} = -l^2$ $\quad \frac{1}{C}\frac{\partial^2 C}{\partial z^2} = p^2 \equiv l^2 + k^2$

X-equation:

- $A(x) = \bar{A}\sin(kx) + B\cos(kx)$

apply BC: $\phi(0) = 0$, $\phi(a) = 0$

- $B = 0$

- $ka = n\pi$

- $k_n = \frac{n\pi}{a}$

* $A(x) = \bar{\bar{A}}\sin\left(\frac{n\pi x}{a}\right)$

y-equation:

$B(y) = \bar{A}\sin(ly) + \bar{B}\cos(ly)$

- apply BC: $\phi(0) = 0$, $\phi(a) = 0$

- $\bar{B} = 0$, $k_m a = m\pi$

$$k = \frac{m\pi}{a}$$

- $B(y) = \bar{A}_m \sin\left(\frac{m\pi x}{a}\right)$

Z-equation:

- $C'' - Cp^2 = 0$

$$C = e^{rz}$$
$$C' = re^{rz}$$
$$C'' = r^2 e^{rz}$$

- $r^2 - p^2 = 0$
- $r = \pm p$

- Real exponentials work, so I'm using combination of hyperbolic sine and cosine.

$$C(z) = A'\sinh(pz) + B'\cosh(pz)$$

apply Boundary condition:

$$\phi(z=0) = 0$$

- $B = 0$ after applying

$$C(z) = A'\sinh(pz)$$

$$\rho_{nm}^2 = l_m^2 + k_n^2 = \frac{n^2\pi^2}{a^2} + \frac{m^2\pi^2}{a^2}$$

$$\rho_{nm} = \pi\sqrt{\frac{n^2+m^2}{a^2}}$$

$$C(z) = A'_{nm}\sinh\left(\pi\sqrt{\frac{n^2+m^2}{a^2}}\,z\right)$$

$$\left(A_{nm} = \bar{\bar{A}}_n A'_{nm} \bar{A}_m\right)$$

$$\Phi(x,y,z) = \sum_{n=1}^{\infty}\sum_{m=1}^{\infty} A_{nm}\sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi x}{a}\right) \sinh\left(\pi\sqrt{\frac{n^2+m^2}{a^2}}\,z\right)$$

- $\int_0^L \sin(nx)\sin(mx) = \frac{L}{2}\delta_{nm}$

apply this twice

apply BC $\quad \phi(z=a) = V_0$

$$\phi(x,y,z\to a) = V_0 = \sum_{n,m=1}^{\infty} A_{nm}\sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi y}{a}\right)\sinh\left(\pi\sqrt{\frac{n^2+m^2}{a^2}}\,a\right)$$

- $V_0 = \sum_{n,m}^{\infty} A_{nm}\sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi y}{a}\right)\sinh\left(\pi\sqrt{n^2+m^2}\right)$

$n' = n$

$m' = m$

$$\frac{4\cdot V_0}{a^2\sinh\left(\pi\sqrt{n^2+m^2}\right)}\int_0^a\int_0^a \sin\left(\frac{n\pi x}{a}\right)\sin\left(\frac{m\pi y}{a}\right)dx\,dy = A_{nm}$$

- $\int_0^a \sin\left(\frac{m\pi y}{a}\right) dy$

$$u = \frac{m\pi y}{a}$$

$$du = \frac{m\pi}{a} dy$$

- $\frac{a}{m\pi} \int_0^{m\pi} \sin(u)\, du$
  - $\frac{2a}{m\pi}$, n = odd
  - 0, n = even
- $\frac{-a}{m\pi} \cos(u) \Big|_0^{m\pi} = \left[\frac{-a}{m\pi}\cos(m\pi) + \frac{a}{m\pi}\right]$

$$\int_0^a \sin\left(\frac{n\pi x}{a}\right) dy = \left[\frac{-a}{n\pi}\cos(n\pi) + \frac{a}{n\pi}\right]$$

$(-1)^n$

- $\frac{2a}{n\pi}$, n = odd
- 0, n = even

$$\Phi(x,y,z) = \sum_{n,m=1,3,5,\ldots}^{\infty} \frac{16 V_0}{\pi^2 nm \sinh\left(\pi\sqrt{n^2+m^2}\right)} \sin\left(\frac{n\pi x}{a}\right) \sin\left(\frac{m\pi y}{a}\right) \sinh\left(\pi \sqrt{\frac{n^2+m^2}{a^2}}\, z\right) \text{ Volts}$$

(f)

- I took the function on the $z=a$ plane
- Charge is distributed in the $z=0$ plane.
- Electric field for charge on $z=0$ plane is the z-direction.
- $\vec{E} = E_z \hat{z}$
- $\vec{E} = -\frac{\partial \Phi}{\partial z}\hat{z}$.

($z=0$ plane)

$$\vec{E} = \frac{\partial \Phi}{\partial z}\hat{z} = \sum_{n,m=\text{odd}}^{\infty} -\left(\frac{n^2\pi^2}{a^2}\frac{16V_0}{nm\pi}\frac{\sinh\left(n\pi x/a\right)\sin\left(m\pi y/a\right)}{\sinh\left(\pi\sqrt{n^2+m^2}\right)}\right)\left(\cosh\left(\pi\sqrt{\frac{n^2+m^2}{a^2}}\,z\right)\right)\hat{z}$$

- $d\vec{A} = dx\,dy\,\hat{z}$

($z=0$ plane)

$0 \le x \le a$

$0 \le y \le a$

- $\int_A \vec{E} \cdot d\vec{A} = \int_A E_z \, dx\, dy = \frac{Q_{Total}}{\varepsilon_0}$

$$\int_0^a \sin\left(\frac{n\pi x}{a}\right) dx \qquad u = \frac{n\pi x}{a}$$

$$du = \frac{n\pi}{a} dx$$

$$\frac{a}{n\pi} \int_0^{n\pi} \sin(u)\, du = -\frac{a}{n\pi} \cos(u) \Big|_0^{n\pi} = \frac{2a}{n\pi}, \quad n = odd$$

$$0, \quad n = even$$

$$\int_0^a \sin\left(\frac{m\pi y}{a}\right) dy = \frac{2a}{m\pi}, \quad m = odd$$

$$0, \quad m = even$$

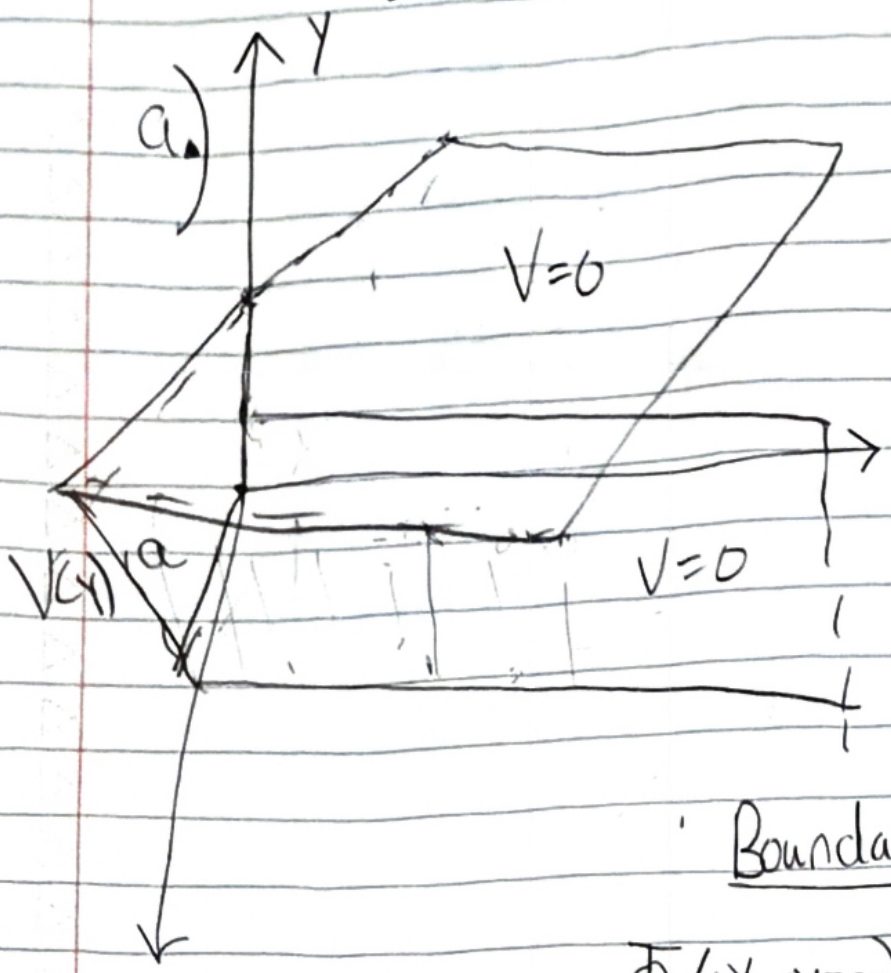
$$Q = -\varepsilon_0 \sum_{n,m=odd}^{\infty} \sqrt{\frac{n^2+m^2}{a^2}} \frac{64 V_0}{n^2 m^2 \pi^4 \sinh\left(\pi \sqrt{n^2+m^2}\right)} \quad (Coulombs)$$

$z = 0$, because

Charge is in $z = 0$ plane

$\cosh(0) = 1$

Prob 3.)



Objective:

a.) Potential between plane.

b.) Surface charge density on lower plate.

Boundary Conditions:

$\Phi(X, Y=a) = 0$

- When x is fixed $\Phi(X, Y=0) = 0$ we need sine and cosine solutions. $\Phi(X=0, Y) = \begin{cases} V_0, & Y < a/2 \\ -V_0, & Y > a/2 \end{cases}$
- When y is fixed $\Phi(X \to \infty, Y) = 0$ we need real exponentials.

(Laplaces equation)

- $\nabla^2 \Phi = 0$
- $\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2}$
- $\frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} = 0$
- $\Phi(x,y) = A(x)B(y)$
- $\frac{1}{A}\frac{\partial^2 A}{\partial x^2} + \frac{1}{B}\frac{\partial^2 B}{\partial y^2} = 0$
- $\frac{1}{A}\frac{\partial^2 A}{\partial x^2} = \frac{-1}{B}\frac{\partial^2 B}{\partial y^2} = k^2$

x-equation:

- $\frac{1}{A}\frac{\partial^2 A}{\partial x^2} = k^2$
- $A'' - Ak^2 = 0$
- $A = e^{rx}$
  $A' = re^{rx}$
  $A'' = r^2 e^{rx}$
- $r^2 - k^2 = 0$

$r = \pm k$,

- $A(x) = Ae^{kx} + Be^{-kx}$

apply boundary conditions.

- $A(x \to \infty) = 0$
- $0 = A\cancel{e^{\infty}} + B(0)$ ← keep this

↑ invalid solution

- $A(x) = B_n e^{-k_n x}$

y-equation:

- $B'' + Bk = 0$
- $B(y) = A\sin(ky) + B\cos(ky)$
- $B = 0$
- $k_n a = n\pi$, $k_n = \frac{n\pi}{a}$

Boundary condition.

$B(a) = 0$

$B(0) = 0$

$B(y) = A_n \sin(k_n y)$

- $A_n = B_n A_n'$

- $\Phi(x,y) = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi y}{a}\right) e^{-\frac{n\pi x}{a}}$

apply B'C:

$$\Phi(x=0,y) = \begin{cases} V_0, & y \le a/2 \\ -V_0, & y \ge \frac{a}{2} \end{cases}$$

For $y \le \frac{a}{2}$:

$$V_0 = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi y}{a}\right)$$

multiply by $\int_{a/2}^{a/2} \sin\left(\frac{m\pi y}{a}\right) dy$

$$V_0 \int_0^{a/2} \sin\left(\frac{m\pi y}{a}\right) dy = \sum_{n=1}^{\infty} A_n \int_0^{a} \sin\left(\frac{m\pi y}{a}\right) \sin\left(\frac{n\pi y}{a}\right) dy \quad (1)$$

For $y \ge a/2$:

$$-V_0 = \sum_{n=1}^{\infty} A_n \sin\left(\frac{n\pi y}{a}\right)$$

multiply by $\int_{a/2}^{a} \sin\left(\frac{m\pi y}{a}\right) dy$

$$-V_0 \int_{a/2}^{a} \sin\left(\frac{m\pi y}{a}\right) dy = \sum_{n=1}^{\infty} A_n \int_{a/2}^{a} \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy$$

(2)

- add relations (1) and (2).

$$V_0 \int_0^{a/2} \sin\left(\frac{m\pi y}{a}\right) dy - V_0 \int_{a/2}^{a} \sin\left(\frac{m\pi y}{a}\right) =$$

$$\sum_{n=1}^{\infty} A_n \left[\int_{a/2}^{a} \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy + \int_0^{a/2} \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy \right]$$

$$\int_0^a \sin\left(\frac{n\pi y}{a}\right) \sin\left(\frac{m\pi y}{a}\right) dy$$

$$V_0 \int_0^{a/2} \sin\left(\frac{n\pi y}{a}\right) dy - V_0 \int_{a/2}^{a} \sin\left(\frac{m\pi y}{a}\right) dy = \frac{a}{2} \delta_{mn} A_n$$

$n = m$

$$\frac{2V_0}{a} \int_0^{a/2} \sin\left(\frac{n\pi y}{a}\right) dy - \frac{2V_0}{a} \int_{a/2}^{a} \sin\left(\frac{n\pi y}{a}\right) dy = A_n .$$

$$\frac{2V_0}{a} \int_0^{a/2} \sin\left(\frac{n\pi y}{a}\right) dy \qquad u = \frac{n\pi y}{a}$$

$$du = \frac{n\pi}{a} dy$$

$$\frac{2V_0}{a}\frac{a}{n\pi}\int_0^{n\pi/2}\sin(v)\,dv$$

$$\frac{2V_0}{n\pi}\left(-\cos(v)\right)_0^{\frac{n\pi}{2}}$$

$$= \frac{2V_0}{n\pi}\left[-\cos\left(\frac{n\pi}{2}\right)+\cos(0)\right]$$

$$= \frac{2V_0}{n\pi}\left[-\cos(n\pi)+1\right]$$

$$= \begin{cases} 2V_0/n\pi & n=\text{odd} \\ 4V_0/n\pi & n=2,6,10,\ldots \\ 0 & n=0,4,8,\ldots \end{cases}$$

$$-\frac{2V_0}{a}\int_{a/2}^{a}\sin\left(\frac{n\pi y}{a}\right)dy \qquad v=\frac{n\pi y}{a}$$

$$dv=\frac{n\pi}{a}dy$$

$$-\frac{2V_0}{n\pi}\int_{\frac{n\pi}{2}}^{n\pi}\sin(u)\,du = \frac{2V_0}{n\pi}\cos(u)\Big|_{\frac{n\pi}{2}}^{n\pi}$$

$$\frac{2V_0}{n\pi}\left(\cos(n\pi) - \cos\left(\frac{n\pi}{2}\right)\right)$$

$$= \begin{cases} -\frac{2V_0}{n\pi} & n = \text{odd} \\ 0 & n = 0, 4, 8, \ldots \\ \frac{4V_0}{n\pi} & n = 2, 6, 10, \ldots \end{cases}$$

Combining Both integrals:

$$A_n = \begin{cases} \frac{8V_0}{n\pi}, & n = 2, 6, 10, \ldots \\ 0, & n = \text{otherwise} \end{cases}$$

$$\Phi(x,y) = \sum_{n=2,6,10\ldots}^{\infty} \frac{8V_0}{n\pi} \sin\left(\frac{n\pi y}{a}\right) e^{-\frac{n\pi x}{a}} \text{ Volts}$$

b.) : lower plane ($y = 0$)
: x-z plane @ $y = 0$.

- Electric field is in the y direction that points downwards @ $y = 0$ plane.

- $\vec{E} = -E_y \hat{y}$

- $\vec{E} = -\frac{\partial \Phi}{\partial y} \hat{y}$

- $\vec{E} = -\frac{\partial}{\partial y}\left(\sum_{n=2,6,10,\ldots}^{\infty} \frac{8V_0}{n\pi} \sin\left(\frac{n\pi y}{a}\right) e^{\frac{-n\pi x}{a}}\right) \hat{y}$

- $\vec{E} = -\sum_{n=2,6,10,\ldots}^{\infty} \frac{8V_0}{n\pi} \frac{n\pi}{a} \cos\left(\frac{n\pi y}{a}\right) e^{\frac{-n\pi x}{a}} \hat{y}$

- $\vec{E} = -\sum_{n=2,6,10,\ldots}^{\infty} \frac{8V_0}{a} \cos\left(\frac{n\pi y}{a}\right) e^{\frac{-n\pi x}{a}} \hat{y}$

$$\vec{\nabla} = \frac{\partial}{\partial y}\hat{y}$$

- $\frac{\rho_s}{\varepsilon_0} = \vec{\nabla} \cdot \vec{E}$

- $\rho_s = -\varepsilon_0 \sum_{n=2,6,10,\ldots}^{\infty} \frac{8V_0}{a}\left(\frac{-n\pi}{a}\right)\sin\left(\frac{n\pi y}{a}\right)e^{\frac{-n\pi x}{a}}$

- $\rho_s = \varepsilon_0 \sum_{n=2,6,10,\ldots}^{\infty} \frac{8V_0 n\pi}{a^2}\sin\left(\frac{n\pi y}{a}\right)e^{-\frac{n\pi x}{a}}$

- for the lower plane $y = 0$

- $\rho_s = 0 \ \frac{C}{m^2}$

Prob 4.)

Objective:

- $R = 1.3 A^{1/3}$ fm
- (Z = 92, A = 235) Uranium nucleus
- $W = \frac{3}{5}\left(\frac{Q^2}{4\pi\varepsilon_0}\right)\frac{1}{R}$

Objective:

a.) Find stored energy in MeV.

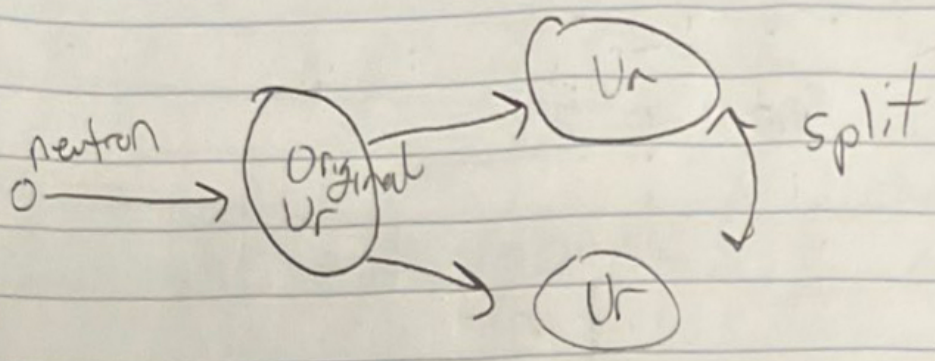
- $\varepsilon_0 = 8.854 \times 10^{-12} \frac{F}{m}$
- $Q = 92(1.6 \times 10^{-19} C) = 1.472 \times 10^{-17} C$
- $R = 1.3(235)^{1/3} = 8.022 \text{ fm}\left(\frac{1 m}{1 \times 10^{15} \text{ fm}}\right) = 8.022 \times 10^{-15} m$
- $1 m = 1.62 \times 10^{-20}$ fm
- $1 J = 6.24 \times 10^{18} eV$

$$W = \frac{3}{5}\left(\frac{(1.472 \times 10^{-17} C)^2}{4\pi(8.854 \times 10^{-12} F/m)}\right)\left(\frac{1}{(8.022 \times 10^{-15} m}\right)$$

$$W = 1.46 \times 10^{-10} J \left(\frac{6.24 \times 10^{18} eV}{1 J}\right)$$

$$W = 9.09 \times 10^{8} eV$$

- $W_{original} = 9.09 \times 10^8 \text{ eV} \left(\frac{1 \text{ MeV}}{1 \times 10^6 \text{ eV}} \right) = 909 \text{ MeV}$



- $2A = 470$
- $R = 1.3(470)^{1/3} \text{ fm} \left(\frac{1 \text{ m}}{1 \times 10^{15} \text{ fm}} \right) = 1.01 \times 10^{-14} \text{ m}$
- $Q = 1.472 \times 10^{-17} \text{ C}$
- $W_{split} = 721.9 \text{ MeV}$
- $\Delta W = W_{original} - W_{split} = 909 \text{ MeV} - 721.9 \text{ MeV}$
  $\approx 188 \text{ MeV}$